

The Sequentially Positive Approximation Property Forces Separability

Automatic Math Discovery Run `fa_banach_001`

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Source And Target

The source paper is arXiv:2605.12117, by Luis A. Garcia, Jose Lucas P. Luiz, and Vinicius C. C. Miranda, on norm attainment for multilinear operators, polynomials, and Banach lattices.

The paper introduces the sequentially positive approximation property (SPAP) for Banach lattices and then asks the following question.

Corollary 4.4. *Let $n \in \mathbb{N}$, E be a reflexive Banach lattice and F be a Dedekind complete Banach lattice. Consider the following implications:*

- (1) *Every $A \in \mathcal{L}^+(E^n; F)$ is norm-attaining.*
- (2) *Every $A \in \mathcal{L}^+(E^n; F)$ is weakly sequentially continuous.*
- (3) *Every $P \in \mathcal{P}^+(E^n; F)$ is weakly sequentially continuous.*
- (4) *Every $P \in \mathcal{P}^+(E^n; F)$ is norm-attaining.*

If E or F has the SPAP, then (1) $\iff$ (2) $\Rightarrow$ (3) $\iff$ (4).

In Section 3, we introduced the SPAP and presented examples of Banach lattices that do and do not have this property. So far, however, the only known class of Banach lattices with the SPAP consists of Banach lattices with order continuous norm whose order is given by a basis. This leads to the following question:

Question 2: Are there other examples of Banach lattices with the SPAP besides Banach lattices with order continuous norm whose order is given by a basis?

We showed in Section 3 that all positive bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ are norm-attaining while there are non-norm attaining bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$. From this and our results, we obtain that all positive bilinear operators $\ell_4 \times \ell_4 \rightarrow L_1([0, 1])$ are weakly sequentially continuous while there are bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ that are not weakly sequentially continuous. It would be interesting, if we get examples with the domain being non-atomic Banach lattices. In view of this, we leave the following question:

Question 3: Are all positive n -linear forms on $L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1]$ with $\sum_{i=1}^n \frac{1}{p_i} < 1$ weakly sequentially continuous?

Question 2 asks whether there are Banach lattices with the SPAP beyond Banach lattices with order-continuous norm whose order is given by a basis.

Definitions

Following the source paper, a Banach lattice E has the *sequentially positive approximation property* if there is a sequence

$$S_n \in B_{\mathcal{F}^+(E; E)}$$

of positive finite-rank contractions such that

$$\| |S_n - \text{id}_E| (x) \| \longrightarrow 0 \quad (x \in E^+),$$

where the modulus $|S_n - \text{id}_E|$ is assumed to exist.

Result

Theorem 1. *If a Banach lattice E has the SPAP, then E is separable.*

Proof. Let (S_n) be an SPAP sequence. Put

$$M = \overline{\text{span}} \bigcup_{n=1}^{\infty} S_n(E).$$

Each $S_n(E)$ is finite dimensional, so M is separable.

We show that every $x \in E$ belongs to M . Since the modulus $|S_n - \text{id}_E|$ exists, the standard lattice inequality for regular operators gives

$$|(S_n - \text{id}_E)x| \leq |S_n - \text{id}_E|(|x|).$$

Taking norms and using the SPAP convergence at the positive vector $|x|$, we obtain

$$\|S_n x - x\| \leq \| |S_n - \text{id}_E| (|x|) \| \longrightarrow 0.$$

But $S_n x \in S_n(E) \subseteq M$ for every n , and M is closed. Thus $x \in M$. Hence $E = M$, so E is separable. $\square$

Corollary 1. *No nonseparable Banach lattice has the SPAP.*

Scope And Limitations

This is a partial negative result for Question 2. It rules out every nonseparable Banach lattice as a possible new example of SPAP, but it does not produce a new example outside the known basis-ordered order-continuous class and does not classify separable Banach lattices.

The result is independent of order continuity, Dedekind completeness, or the presence of a weak unit, except for the source definition's requirement that the moduli $|S_n - \text{id}_E|$ exist.

Novelty And Duplicate Check

Local duplicate check: the run registry and lightweight indexes were searched for arXiv:2605.12117, SPAP, separability, and nonseparable Banach lattices. Existing packets cover non-atomic finite-measure L_p spaces, infinite-dimensional AM-spaces with a strong unit, and Question 3 on positive multilinear forms, but not this separability obstruction.

Source check: the source text around the definition of SPAP and Question 2 was inspected. No same-paper statement of Theorem 1 was found.

External search on 2026-06-13 used exact phrases around the sequentially positive approximation property, the abbreviation SPAP, separability, nonseparable Banach lattices, and the source-paper title. The only relevant hit found was the source arXiv page itself.

Human Review Recommendation

Check only the standard inequality

$$|Tx| \leq |T|(|x|)$$

for regular operators with existing modulus. Once this is accepted, the separability argument is immediate from the countable sequence of finite-rank ranges.